Lecture Scribe Joint Variables

L15_L17.pdf
PDF

Generate a lecture scribe intended to serve as faithful, exam-ready reference material for revision.

Use only the provided context consisting of the lecture slides and relevant textbook excerpts. Do not introduce any material that does not appear in the given context.

While writing the scribe, reason step by step through all definitions, notation, assumptions, theorems or results, proofs or derivations, and worked example solutions exactly as presented in the context.

Make the logical flow between concepts explicit so that a student can clearly follow the reasoning during exam preparation.

Maintain a formal academic tone and structure the scribe clearly for revision purposes.

Below is a **lecture scribe created strictly from the provided slides**. No external material has been added. The explanation follows the **same logical sequence as the lecture** and expands the reasoning between steps to make it **exam-ready revision material**.

Source:

Lecture Scribe

Joint Random Variables and Joint Distributions

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1. Motivation: Why Study Joint Random Variables?

In many scientific and real-world problems, we are interested in **two or more random variables simultaneously** rather than analyzing them individually. Studying them together allows us to understand the **relationship or dependence** between the variables.

Examples of Paired Random Variables

Typical examples include:

- Height and weight of giraffes
- IQ and birthweight of children
- Frequency of exercise and rate of heart disease
- Air pollution level and respiratory illness rate
- Number of Facebook friends and age of members

These examples illustrate situations where **two variables interact or influence each other**.

Engineering and Computing Applications

Several practical applications illustrate the importance of studying joint random variables.

Example 1: Smart Transportation
Two random variables are considered:

$$X = \text{Vehicle Speed}$$

$$Y = \text{Traffic Density}$$

Key question:

> Can vehicle speed be high when traffic density is also high?

This analysis helps in:

- Adaptive traffic signal systems
- Congestion prediction
- Autonomous vehicle routing

Because the values of X and Y influence each other, they are **correlated random variables**.

Example 2: Wireless Network Performance
Two important variables are:

$$X = \text{Signal-to-Noise Ratio (SNR)}$$

$$Y = \text{Data Throughput}$$

Throughput often depends on SNR. However, even with high SNR, throughput may not always be high due to **network congestion**. This illustrates **conditional relationships between variables**.

Applications include:

- 5G scheduling
- Adaptive modulation
- Network planning

Additional Examples

- Weather prediction:
  $X =$ Rainfall, $Y =$ Temperature
- Drone delivery:
  $X =$ Wind speed, $Y =$ Delivery time
- Dice experiment:
  Two dice outcomes analyzed jointly

These examples demonstrate that **many real-world problems require analyzing multiple random variables simultaneously**.

Role of Joint Distributions

When random variables appear together, they follow a **joint distribution**.

The joint distribution allows us to:

1. Compute probabilities involving multiple variables
2. Understand relationships between variables

If the variables are **independent**, analysis becomes simpler. Otherwise, dependence is analyzed using measures such as:

- Covariance
- Correlation

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2. Discrete Case: Joint Probability Mass Function (Joint PMF)

Setup

Suppose two discrete random variables:

$$X = \{x_1, x_2, ..., x_n\}$$

$$Y = \{y_1, y_2, ..., y_m\}$$

The ordered pair (X, Y) takes values from the **product set**:

$$\{(x_1, y_1), (x_1, y_2), ..., (x_n, y_m)\}$$

Each element represents a **joint outcome**.

Definition: Joint PMF

The **joint probability mass function** is defined as:

$$p(x_i, y_j) = P(X = x_i, Y = y_j)$$

This represents the probability that:

- X equals x_i, and
- Y equals y_j

simultaneously.

Joint PMF Table Representation

The joint PMF is often organized as a **table** where:

- Rows correspond to values of X
- Columns correspond to values of Y

Each entry contains:

$$p(x_i, y_j)$$

Key Properties

1. Probability values lie between 0 and 1

$$0 \leq p(x_i, y_j) \leq 1$$

2. Total probability equals 1

$$\sum_{i=1}^{n} \sum_{j=1}^{m} p(x_i, y_j) = 1$$

These conditions ensure the function is a **valid probability distribution**.

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3. Example: Rolling Two Dice

Experiment

Two fair dice are rolled.

Define:

$$X = \text{value on the first die}$$

$$T = \text{total (sum) of both dice}$$

Each pair (i, j) occurs with probability:

$$\frac{1}{36}$$

since there are 36 equally likely outcomes.

Joint Probability Table

The joint table for (X, T) lists probabilities for combinations such as:

$$(X = 1, T = 2), (X = 1, T = 3), ..., (X = 6, T = 12)$$

Entries in the table are either:

$$0 \quad \text{or} \quad \frac{1}{36}$$

depending on whether the combination is possible.

Observation

The variables X and T are **not independent**.

Reason:

- The value of T depends on the value of X.
- If $X = 6$, then T cannot be less than 7.

Thus the variables are **dependent random variables**.

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4. Continuous Case: Joint Probability Density Function (Joint PDF)

The continuous case is **analogous to the discrete case**, with the following replacements:

Discrete	Continuous
Discrete values	Continuous intervals
Joint PMF	Joint PDF
Summation	Double integral

Product Region

If:

$$X \in [a, b], \quad Y \in [c, d]$$

then the pair (X, Y) lies in the **product region**

$$[a, b] \times [c, d]$$

This region represents all possible joint values.

Definition: Joint PDF

A function $f(x, y)$ is the **joint probability density function** if:

$$P((X, Y) \text{ lies in a small rectangle}) \approx f(x, y)\, dx\, dy$$

where the rectangle has area:

$$dx \times dy$$

Thus:

$$\text{Probability} \approx f(x, y)dxdy$$

This is illustrated geometrically as a small rectangle in the (x, y) plane.

Key Properties of Joint PDF

1. Non-negativity

$$f(x, y) \geq 0$$

2. Total probability equals 1

$$\int_c^d \int_a^b f(x, y)\, dx\, dy = 1$$

Important note:

The value of $f(x, y)$ **may exceed 1** because it is a **density**, not a probability.

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5. Worked Example (Continuous Joint PDF)

Given

$$f(x, y) = 4xy, \quad 0 \leq x \leq 1, \quad 0 \leq y \leq 1$$

Thus (X, Y) lies in the **unit square**

$$[0, 1]^2$$

Tasks:

1. Verify that $f(x, y)$ is a valid joint PDF
2. Visualize the event

$$A = \{X < 0.5, Y > 0.5\}$$

3. Compute $P(A)$.

Step 1: Check Validity (Normalization)

Compute total probability:

$$\int_0^1 \int_0^1 4xy \, dx \, dy$$

First integrate with respect to x:

$$\int_0^1 4xy \, dx = 4y \left[\frac{x^2}{2} \right]_0^1$$

$$= 2y$$

Now integrate with respect to y:

$$\int_0^1 2y \, dy$$

$$= [y^2]_0^1$$

$$= 1$$

Since:

- $f(x, y) \geq 0$
- total probability = 1

the function is a **valid joint PDF**.

Step 2: Compute $P(A)$

Event:

$$A = \{X < 0.5, Y > 0.5\}$$

Thus:

$$P(A) = \int_0^{0.5} \int_{0.5}^1 4xy \, dy \, dx$$

First integrate with respect to y:

$$\int_{0.5}^{1} 4xy\,dy$$

$$= 4x\left[\frac{y^2}{2}\right]_{0.5}^{1}$$

$$= 2x(1-0.25)$$

$$= \frac{3}{2}x$$

Now integrate with respect to x:

$$\int_{0}^{0.5} \frac{3}{2}x\,dx$$

$$= \frac{3}{2}\left[\frac{x^2}{2}\right]_{0}^{0.5}$$

$$= \frac{3}{16}$$

Therefore:

$$P(A) = \frac{3}{16}$$

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6. Joint Cumulative Distribution Function (Joint CDF)

Suppose X and Y are jointly distributed random variables.

The notation:

$$X \leq x, \quad Y \leq y$$

represents the event:

$$\{X \leq x \text{ and } Y \leq y\}$$

Definition

The **joint cumulative distribution function** is defined as:

$$F(x, y) = P(X \leq x, Y \leq y)$$

This represents the probability that both random variables are **less than or equal to specified values simultaneously**.

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7. Joint CDF for Continuous Variables

If X and Y have joint density $f(x, y)$ over region $[a, b] \times [c, d]$, then

$$F(x, y) = \int_c^y \int_a^x f(u, v)\, du\, dv$$

This means the CDF accumulates probability by **integrating the density over the rectangle** from the lower bounds to (x, y).

Recovering the Joint PDF

The joint density can be obtained from the CDF using partial derivatives:

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

Thus the CDF and PDF are mathematically related through differentiation and integration.

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8. Joint CDF for Discrete Variables

If X and Y are discrete with joint PMF $p(x_i, y_j)$, then

$$F(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(x_i, y_j)$$

Interpretation:

To compute $F(x, y)$:

1. Select all rows with $x_i \leq x$
2. Select all columns with $y_j \leq y$
3. Add all probabilities in that region of the table.

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9. Properties of the Joint CDF

The joint CDF $F(x, y)$ satisfies several important properties.

1. Non-decreasing property

$$F(x, y) \text{ is non-decreasing}$$

If x or y increases, $F(x, y)$ either:

- stays the same, or
- increases.

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2. Lower-left boundary condition

$$F(x, y) = 0$$

at the lower-left corner of the joint range.

If the range extends to $(-\infty, -\infty)$:

$$\lim_{(x,y)\to(-\infty,-\infty)} F(x, y) = 0$$

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3. Upper-right boundary condition

$$F(x, y) = 1$$

at the upper-right corner.

If the range extends to (∞, ∞):

$$\lim_{(x,y)\to(\infty,\infty)} F(x, y) = 1$$

These properties guarantee that the function behaves like a **valid cumulative probability function**.

Final Conceptual Summary

The lecture establishes the framework for analyzing **multiple random variables simultaneously**.

Key ideas:

1. Joint random variables model relationships between quantities.
2. For **discrete variables**, probabilities are described using the **joint PMF**.
3. For **continuous variables**, probabilities are described using the **joint PDF** and computed via **double integrals**.
4. The **joint CDF** represents cumulative probability and connects to the PDF through differentiation.
5. Dependence between variables can be observed through the joint distribution.

If you want, I can also convert this into a **2–3 page exam cheat sheet (all formulas + steps + typical questions)** which is extremely useful for **probability exams like yours.**